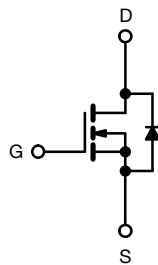
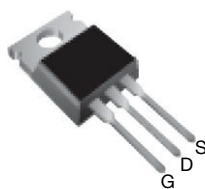


# Power MOSFET

**TO-220AB**


N-Channel MOSFET

## FEATURES

- Dynamic dV/dt rating
- Logic-level gate drive
- $R_{DS(on)}$  specified at  $V_{GS} = 4\text{ V}$  and  $5\text{ V}$
- $175\text{ }^{\circ}\text{C}$  operating temperature
- Fast switching
- Ease of paralleling
- Simple drive requirements
- Material categorization: for definitions of compliance please see [www.vishay.com/doc?99912](http://www.vishay.com/doc?99912)


**RoHS\***  
Available

## Note

\* This datasheet provides information about parts that are RoHS-compliant and / or parts that are non RoHS-compliant. For example, parts with lead (Pb) terminations are not RoHS-compliant. Please see the information / tables in this datasheet for details

## DESCRIPTION

Third generation power MOSFETs from Vishay provide the designer with the best combination of fast switching, ruggedized device design, low on-resistance and cost-effectiveness.

The TO-220AB package is universally preferred for all commercial-industrial applications at power dissipation levels to approximately 50 W. The low thermal resistance and low package cost of the TO-220AB contribute to its wide acceptance throughout the industry.

## PRODUCT SUMMARY

|                           |                         |       |
|---------------------------|-------------------------|-------|
| $V_{DS}$ (V)              | 60                      |       |
| $R_{DS(on)}$ ( $\Omega$ ) | $V_{GS} = 5.0\text{ V}$ | 0.028 |
| $Q_g$ (Max.) (nC)         | 66                      |       |
| $Q_{gs}$ (nC)             | 12                      |       |
| $Q_{gd}$ (nC)             | 43                      |       |
| Configuration             | Single                  |       |

## ORDERING INFORMATION

|                                 |               |
|---------------------------------|---------------|
| Package                         | TO-220AB      |
| Lead (Pb)-free                  | IRLZ44PbF     |
| Lead (Pb)-free and halogen-free | IRLZ44PbF-BE3 |

## ABSOLUTE MAXIMUM RATINGS ( $T_C = 25\text{ }^{\circ}\text{C}$ , unless otherwise noted)

| PARAMETER                                                 |                        |                         | SYMBOL                            | LIMIT       | UNIT     |
|-----------------------------------------------------------|------------------------|-------------------------|-----------------------------------|-------------|----------|
| Drain-source voltage                                      |                        |                         | V <sub>DS</sub>                   | 60          | V        |
| Gate-source voltage                                       |                        |                         | V <sub>GS</sub>                   | ± 10        |          |
| Continuous drain current                                  | V <sub>GS</sub> at 5 V | T <sub>C</sub> = 25 °C  | I <sub>D</sub>                    | 50          | A        |
|                                                           |                        | T <sub>C</sub> = 100 °C |                                   | 36          |          |
| Pulsed drain current <sup>a</sup>                         |                        |                         | I <sub>DM</sub>                   | 200         |          |
| Linear derating factor                                    |                        |                         |                                   | 1.0         | W/°C     |
| Single pulse avalanche energy <sup>b</sup>                |                        |                         | E <sub>AS</sub>                   | 400         | mJ       |
| Maximum power dissipation                                 | T <sub>C</sub> = 25 °C |                         | P <sub>D</sub>                    | 150         | W        |
| Peak diode recovery dV/dt <sup>c</sup>                    |                        |                         | dV/dt                             | 4.5         | V/ns     |
| Operating junction and storage temperature range          |                        |                         | T <sub>J</sub> , T <sub>stg</sub> | -55 to +175 | °C       |
| Soldering recommendations (peak temperature) <sup>d</sup> | For 10 s               |                         |                                   | 300         |          |
| Mounting torque                                           | 6-32 or M3 screw       |                         |                                   | 10          | lbf · in |
|                                                           |                        |                         |                                   | 1.1         | N · m    |

## Notes

- Repetitive rating; pulse width limited by maximum junction temperature (see fig. 11)
- $V_{DD} = 25\text{ V}$ , starting  $T_J = 25\text{ }^{\circ}\text{C}$ ,  $L = 179\text{ }\mu\text{H}$ ,  $R_g = 25\text{ }\Omega$ ,  $I_{AS} = 51\text{ A}$  (see fig. 12)
- $I_{SD} \leq 51\text{ A}$ ,  $dV/dt \leq 250\text{ A/s}$ ,  $V_{DD} \leq V_{DS}$ ,  $T_J \leq 175\text{ }^{\circ}\text{C}$
- 1.6 mm from case
- Current limited by the package, (die current = 51 A)

**THERMAL RESISTANCE RATINGS**

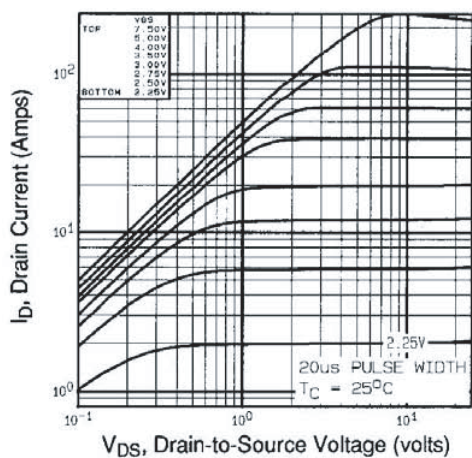
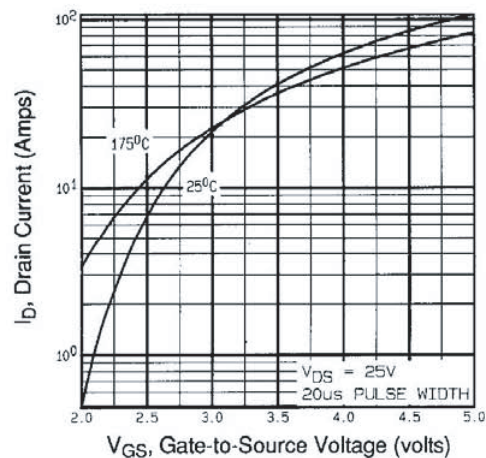
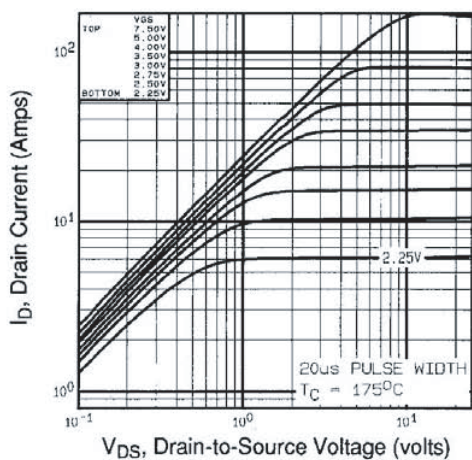
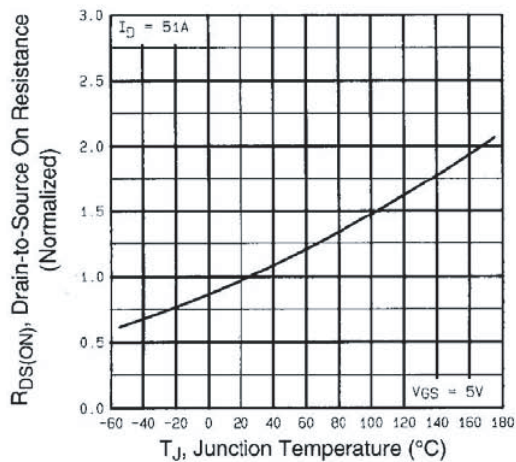
| PARAMETER                           | SYMBOL     | TYP. | MAX. | UNIT |
|-------------------------------------|------------|------|------|------|
| Maximum junction-to-ambient         | $R_{thJA}$ | -    | 62   | °C/W |
| Case-to-sink, flat, greased surface | $R_{thCS}$ | 0.50 | -    |      |
| Maximum junction-to-case (drain)    | $R_{thJC}$ | -    | 1.0  |      |

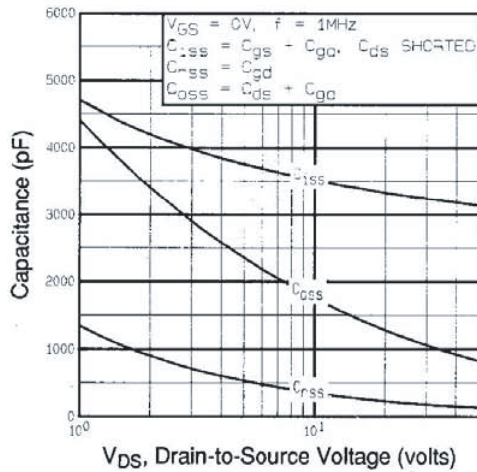
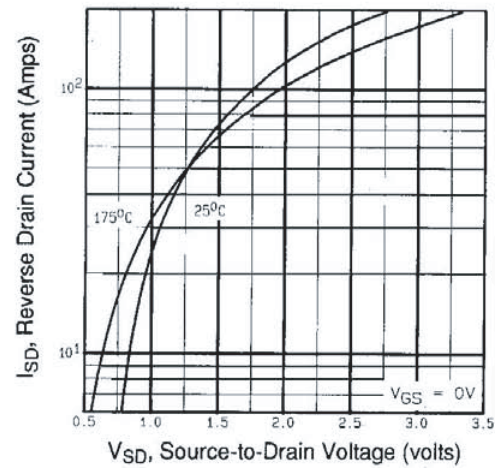
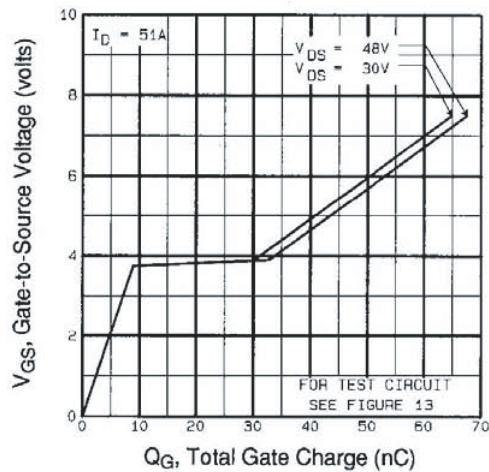
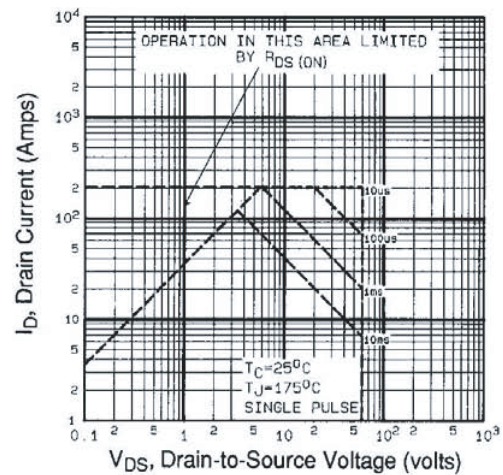
**SPECIFICATIONS** ( $T_J = 25\text{ }^{\circ}\text{C}$ , unless otherwise noted)

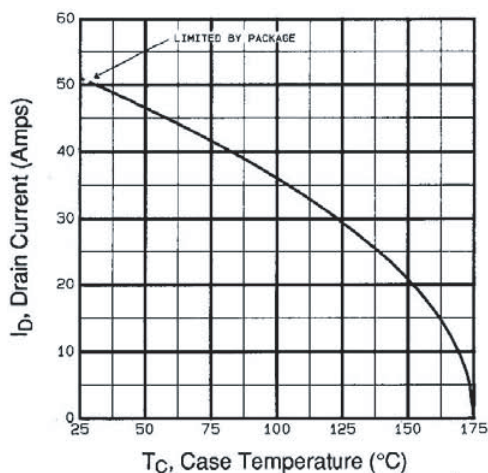
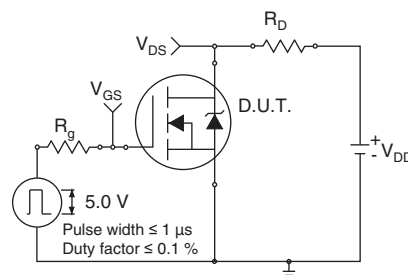
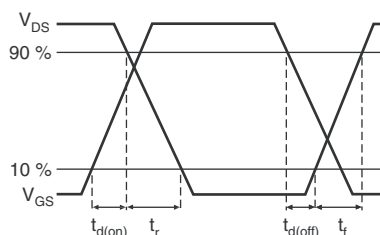
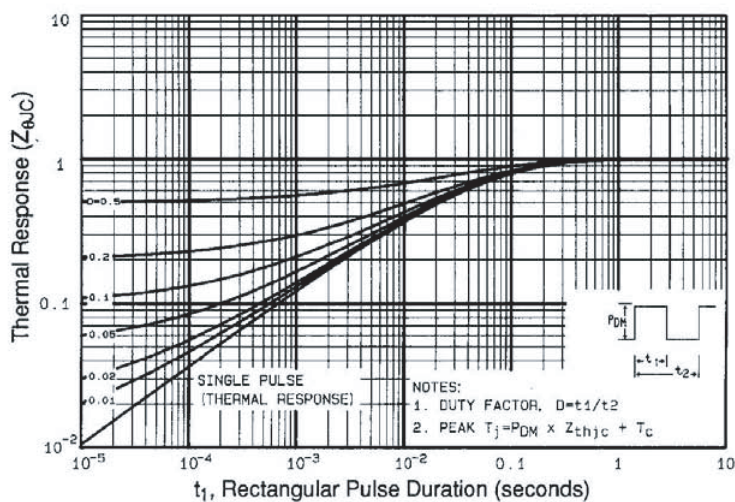
| PARAMETER                                 | SYMBOL                           | TEST CONDITIONS                                                                                                             |                                                                                  | MIN. | TYP.  | MAX.            | UNIT |
|-------------------------------------------|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|------|-------|-----------------|------|
| Static                                    |                                  |                                                                                                                             |                                                                                  |      |       |                 |      |
| Drain-source breakdown voltage            | V <sub>DS</sub>                  | V <sub>GS</sub> = 0 V, I <sub>D</sub> = 250 μA                                                                              |                                                                                  | 60   | -     | -               | V    |
| V <sub>DS</sub> temperature coefficient   | ΔV <sub>DS</sub> /T <sub>J</sub> | Reference to 25 °C, I <sub>D</sub> = 1 mA                                                                                   |                                                                                  | -    | 0.070 | -               | V/°C |
| Gate-source threshold voltage             | V <sub>GS(th)</sub>              | V <sub>DS</sub> = V <sub>GS</sub> , I <sub>D</sub> = 250 μA                                                                 |                                                                                  | 1.0  | -     | 2.0             | V    |
| Gate-source leakage                       | I <sub>GSS</sub>                 | V <sub>GS</sub> = 10 V                                                                                                      |                                                                                  | -    | -     | ± 100           | nA   |
| Zero gate voltage drain current           | I <sub>DSS</sub>                 | V <sub>DS</sub> = 60 V, V <sub>GS</sub> = 0 V                                                                               |                                                                                  | -    | -     | 25              | μA   |
|                                           |                                  | V <sub>DS</sub> = 48 V, V <sub>GS</sub> = 0 V, T <sub>J</sub> = 150 °C                                                      |                                                                                  | -    | -     | 250             |      |
| Drain-source on-state resistance          | R <sub>DS(on)</sub>              | V <sub>GS</sub> = 5.0 V                                                                                                     | I <sub>D</sub> = 31 A <sup>b</sup>                                               | -    | -     | 0.028           | Ω    |
|                                           |                                  | V <sub>GS</sub> = 4.0 V                                                                                                     | I <sub>D</sub> = 25 A <sup>b</sup>                                               | -    | -     | 0.039           |      |
| Forward transconductance                  | g <sub>fs</sub>                  | V <sub>DS</sub> = 25 V, I <sub>D</sub> = 31 A <sup>b</sup>                                                                  |                                                                                  | 23   | -     | -               | S    |
| Dynamic                                   |                                  |                                                                                                                             |                                                                                  |      |       |                 |      |
| Input capacitance                         | C <sub>iss</sub>                 | V <sub>GS</sub> = 0 V,<br>V <sub>DS</sub> = 25 V,<br>f = 1.0 MHz, see fig. 5                                                |                                                                                  | -    | 3300  | -               | pF   |
| Output capacitance                        | C <sub>oss</sub>                 |                                                                                                                             |                                                                                  | -    | 1200  | -               |      |
| Reverse transfer capacitance              | C <sub>rss</sub>                 |                                                                                                                             |                                                                                  | -    | 200   | -               |      |
| Total gate charge                         | Q <sub>g</sub>                   | V <sub>GS</sub> = 5.0 V                                                                                                     | I <sub>D</sub> = 51 A, V <sub>DS</sub> = 48 V,<br>see fig. 6 and 13 <sup>b</sup> | -    | -     | 66              | nC   |
| Gate-source charge                        | Q <sub>gs</sub>                  |                                                                                                                             |                                                                                  | -    | -     | 12              |      |
| Gate-drain charge                         | Q <sub>gd</sub>                  |                                                                                                                             |                                                                                  | -    | -     | 43              |      |
| Turn-on delay time                        | t <sub>d(on)</sub>               | V <sub>DD</sub> = 30 V, I <sub>D</sub> = 51 A,<br>R <sub>g</sub> = 4.6 Ω, R <sub>D</sub> = 0.56 Ω, see fig. 10 <sup>b</sup> |                                                                                  | -    | 17    | -               | ns   |
| Rise time                                 | t <sub>r</sub>                   |                                                                                                                             |                                                                                  | -    | 230   | -               |      |
| Turn-off delay time                       | t <sub>d(off)</sub>              |                                                                                                                             |                                                                                  | -    | 42    | -               |      |
| Fall time                                 | t <sub>f</sub>                   |                                                                                                                             |                                                                                  | -    | 110   | -               |      |
| Internal drain inductance                 | L <sub>D</sub>                   | Between lead,<br>6 mm (0.25") from<br>package and center of<br>die contact                                                  |                                                                                  | -    | 4.5   | -               | nH   |
| Internal source inductance                | L <sub>S</sub>                   |                                                                                                                             |                                                                                  | -    | 7.5   | -               |      |
| Drain-Source Body Diode Characteristics   |                                  |                                                                                                                             |                                                                                  |      |       |                 |      |
| Continuous source-drain diode current     | I <sub>S</sub>                   | MOSFET symbol<br>showing the<br>integral reverse<br>p - n junction diode                                                    |                                                                                  | -    | -     | 50 <sup>c</sup> | A    |
| Pulsed diode forward current <sup>a</sup> | I <sub>SM</sub>                  |                                                                                                                             |                                                                                  | -    | -     | 200             |      |
| Body diode voltage                        | V <sub>SD</sub>                  | T <sub>J</sub> = 25 °C, I <sub>S</sub> = 51 A, V <sub>GS</sub> = 0 V <sup>b</sup>                                           |                                                                                  | -    | -     | 2.5             | V    |
| Body diode reverse recovery time          | t <sub>rr</sub>                  | T <sub>J</sub> = 25 °C, I <sub>F</sub> = 51 A, dI/dt = 100 A/μs <sup>b</sup>                                                |                                                                                  | -    | 130   | 180             | ns   |
| Body diode reverse recovery charge        | Q <sub>rr</sub>                  |                                                                                                                             |                                                                                  | -    | 0.84  | 1.3             | μC   |
| Forward turn-on time                      | t <sub>on</sub>                  | Intrinsic turn-on time is negligible (turn-on is dominated by L <sub>S</sub> and L <sub>D</sub> )                           |                                                                                  |      |       |                 |      |

**Notes**

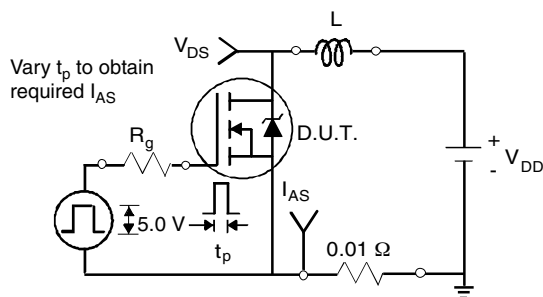
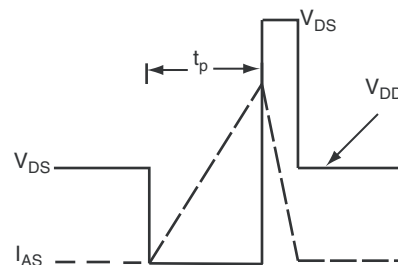
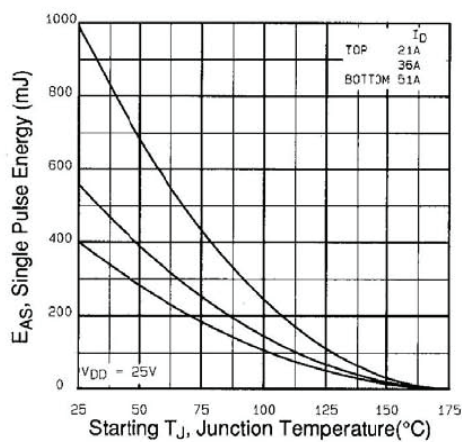
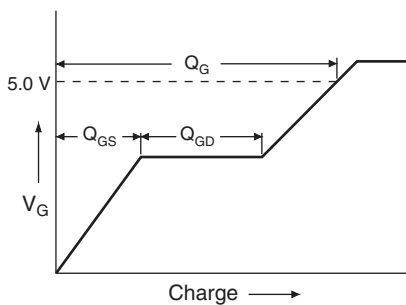
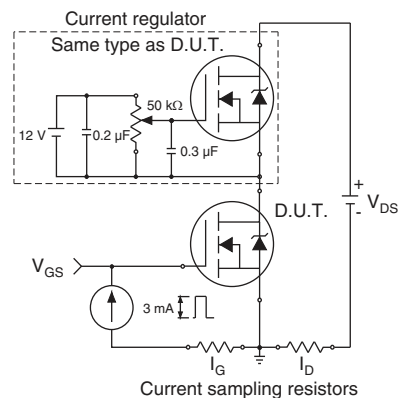
- Repetitive rating; pulse width limited by maximum junction temperature (see fig. 11)
- Pulse width  $\leq 300\text{ }\mu\text{s}$ ; duty cycle  $\leq 2\%$
- Current limited by the package, (die current = 51 A)

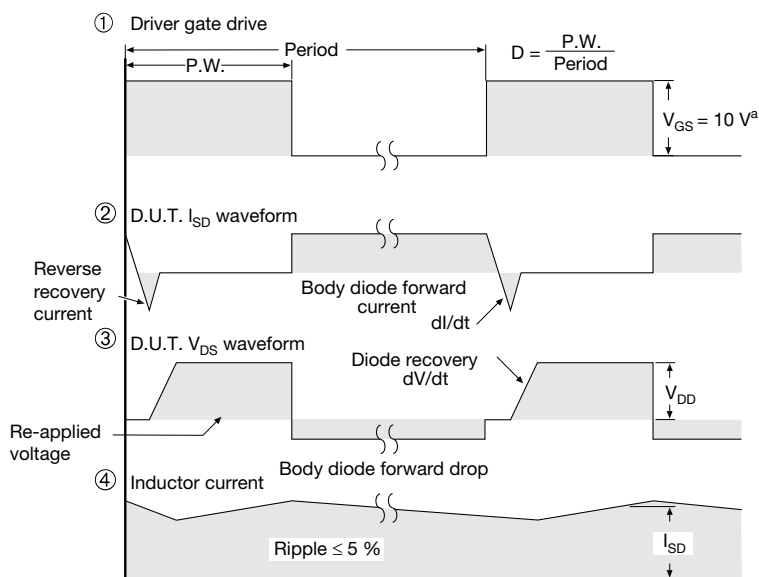
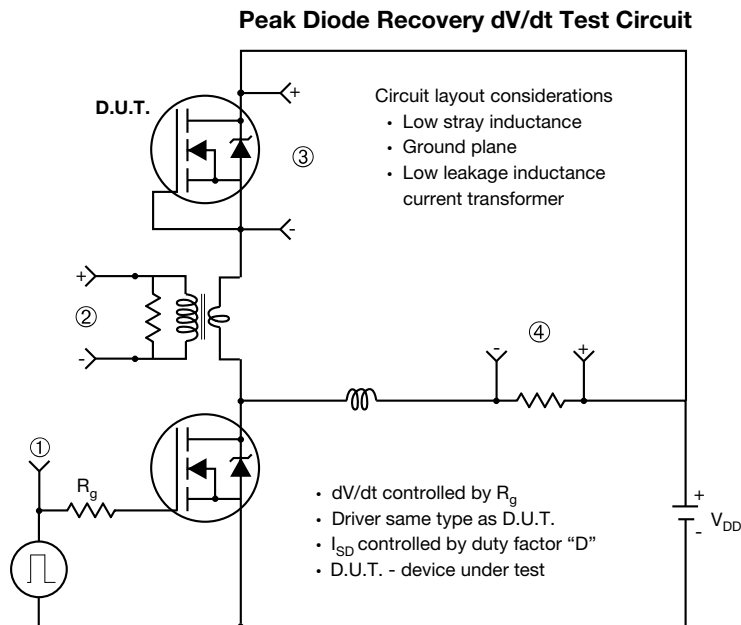
**TYPICAL CHARACTERISTICS** (25 °C, unless otherwise noted)

**Fig. 1 - Typical Output Characteristics,  $T_C = 25^\circ\text{C}$** 

**Fig. 3 - Typical Transfer Characteristics**

**Fig. 2 - Typical Output Characteristics,  $T_C = 175^\circ\text{C}$** 

**Fig. 4 - Normalized On-Resistance vs. Temperature**


**Fig. 5 - Typical Capacitance vs. Drain-to-Source Voltage**

**Fig. 7 - Typical Source-Drain Diode Forward Voltage**

**Fig. 6 - Typical Gate Charge vs. Gate-to-Source Voltage**

**Fig. 8 - Maximum Safe Operating Area**


**Fig. 9 - Maximum Drain Current vs. Case Temperature**

**Fig. 10a - Switching Time Test Circuit**

**Fig. 10b - Switching Time Waveforms**

**Fig. 10 - Maximum Effective Transient Thermal Impedance, Junction-to-Case**




**Fig. 12a - Unclamped Inductive Test Circuit**

**Fig. 12b - Unclamped Inductive Waveforms**

**Fig. 12c - Maximum Avalanche Energy vs. Drain Current**

**Fig. 13a - Basic Gate Charge Waveform**

**Fig. 13b - Gate Charge Test Circuit**


**Note**

a.  $V_{GS} = 5 V$  for logic level devices

**Fig. 11 - For N-Channel**

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